LoSpar-CI

Progressive Low-Rank-plus-Sparse Tensor Inference

Communicating intrinsic structure — not dense tensors — across heterogeneous collaborating devices

Collaborative inference for ultra-high-dimensional data should communicate *intrinsic structure* — low-rank factors and sparse innovations — rather than dense intermediate tensors. A device sends a coarse low-rank approximation first, and transmits sparse residual blocks only if the downstream prediction is not *certifiably* stable.

This document gives the motivation, the related work with references (low-rank+sparse foundations, token compression, successive refinement, compressed sensing), the method with its certified early-stop rule and communication analysis, the application domains, the theory targets, and a scoped first-paper evaluation.

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1 Motivation

Motivation in one paragraph

When several devices collaborate to run inference on ultra-high-dimensional inputs — gigapixel whole-slide pathology images, hyperspectral cubes, 3D medical volumes, long video — the intermediate tensors that must cross device boundaries are enormous, yet they are *highly structured*: most of their energy lies in a low-rank component, with a small number of rare but informative sparse events on top, and negligible residual noise. Sending these tensors *densely* wastes the scarce inter-device link. LoSpar-CI instead communicates the *structure*: a coarse low-rank part first, then sparse residual blocks *only when* a certificate says the coarse answer is not good enough. The result is communication that scales with intrinsic complexity, not ambient size, with *no* uncertified accuracy loss.

1.1 Why the dense status quo is wasteful

Collaborative-inference systems [1, 2, 3] transmit dense activations between stages. For a hidden tensor of ambient size $N = \prod_d n_d$ this cost grows with N. But for the target modalities, the effective degrees of freedom are far smaller: hyperspectral cubes are canonically low-rank plus sparse noise [37, 38]; dynamic medical volumes decompose into low-rank plus sparse components [40]; whole-slide images carry extreme spatial redundancy [43]. Communicating N numbers to convey a few hundred effective ones is the inefficiency LoSpar-CI removes.

1.2 Why "compress before inference" is not enough

Token-compression methods (§2.2) prune or merge tokens *inside one model's forward pass*, on a schedule fixed at design time by attention heuristics, with no downstream guarantee. LoSpar-CI is different in three ways: (i) it communicates a *decomposition* rather than a pruned token set; (ii) refinement is *demand-driven* by a data-dependent certificate; and (iii) it spans *multiple devices* under a communication budget.

2 Background and related work

2.1 Low-rank-plus-sparse decomposition

The $A = L + S$ model originates in **Robust PCA / Principal Component Pursuit** [4]: recover a low-rank L and sparse S by minimizing $\|L\|_* + \lambda\|S\|_1$. Its tensor generalization is **Tensor Robust PCA** [5, 6] using the t-SVD tensor nuclear norm. The low-rank part can be encoded with the classical **CP / Tucker** decompositions [7] or the **Tensor-Train** [8] format, whose factor sizes — not the ambient size — determine storage and communication. In neural networks, **LoSparse** [9] compresses weight matrices as low-rank + sparse ($W \approx UV + S$), and **Linformer** [10] shows attention matrices are approximately low-rank — direct evidence that the intermediate objects LoSpar-CI transmits carry a strong L term.

2.2 Token compression for large visual inputs

The nearest "compress inside inference" line: **DTC-WSI** [25] dynamically merges redundant tokens for gigapixel whole-slide images (reported ∼40% of tokens retained, ∼5–10× compute reduction); **SparseVLM** [26] is a training-free, text-guided visual-token sparsifier (∼54% FLOPs cut, ∼97% accuracy retained); **ToMe** [27] merges similar tokens by bipartite soft matching; **FastV** [28] drops ∼50% of visual tokens after a shallow layer; **VoCo-LLaMA** [29] and **LLaVA-PruMerge** [30] distill/prune tokens to a small retained set. All reduce tokens for a *single* model on *one* device with a *fixed* schedule and *no* downstream certificate.

2.3 Progressive / successive-refinement transmission

Sending coarse-then-fine has deep roots: **successive refinement of information** [31] characterizes when a source can be encoded in layers that are each rate-distortion optimal (progressive JPEG [32] is the engineering analog). LoSpar-CI's *certified early stop* is the inference-side analog of **early-exit** networks — BranchyNet [33] and DeeBERT [34] halt when a confidence criterion is met. The novelty here is stopping based on a *reconstruction-error / output-margin certificate* tied to the transmitted decomposition, across devices.

2.4 Compressed-sensing communication bound

Recovering a k-sparse vector in $\mathbb{R}^N$ needs only $m = O(k \log(N/k))$ random linear measurements via ℓ_1 minimization [35, 36] — the result that underwrites LoSpar-CI's claim that transmitting the sparse residual S costs $O(k \log(N/k))$, not $O(N)$.

3 Method

3.1 The structural model

At a chosen pipeline cut, the intermediate tensor $A_\ell \in \mathbb{R}^N$ at layer/stage ℓ is modeled as

$$A_\ell = \underbrace{L_\ell}_{\text{low-rank (shared structure)}} + \underbrace{S_\ell}_{\text{sparse (rare informative events)}} + \underbrace{E_\ell}_{\text{small residual noise}},$$

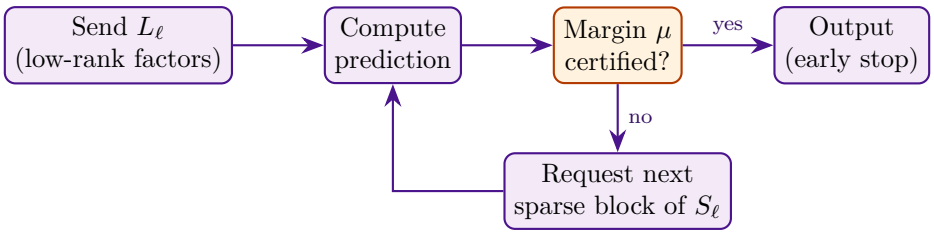
with $\operatorname{rank}(L_\ell) \le r$, $\|S_\ell\|_0 \le k$, and $\|E_\ell\| \le \varepsilon_0$.

3.2 Encoding the two structured parts

- L_ℓ **via factors:** a Tensor-Train (or Tucker/CP) representation [8, 7]; transmit the factors, of total size $\ll N$.
- S_ℓ **via support + values:** the index set $\Omega = \operatorname{supp}(S_\ell)$ (as block indices or a bitmap) plus the k nonzero values; cost $O(k \log(N/k))$ by the compressed-sensing bound [35, 36].

3.3 The progressive protocol

1. **Send coarse first.** The sending device transmits only the low-rank factors of L_ℓ (a coarse approximation $\hat{A}_\ell^{(0)} = L_\ell$).
2. **Compute and certify.** The receiving device(s) run the downstream prediction on $\hat{A}_\ell^{(0)}$ and compute an output *margin* μ (e.g. top-1 minus top-2 logit, or a calibrated uncertainty).
3. **Refine only if needed.** If the margin does not certify stability (§3.4), request additional sparse residual blocks of S_ℓ — largest-energy blocks first — and recompute; repeat until certified or S_ℓ is exhausted.
4. **Joint selection.** Choose the rank r, the sparse budget k, the pipeline cut ℓ, and the target device *together*, as HexGen chooses placement [1] — here trading reconstruction fidelity against communication and compute.



3.4 The certified early-stop rule

Let F be the downstream map from the cut to the output, and suppose F is κ-Lipschitz (local Lipschitz constants are estimable per layer). If the current reconstruction $\hat{A}_\ell = L_\ell + S_\ell^{(j)}$ (after j residual blocks) satisfies $\|A_\ell - \hat{A}_\ell\| \le \eta_j$, then the output moves by at most $\kappa\eta_j$.

Theory target

Certified equivalence. If the output margin on the coarse prediction exceeds the worst-case perturbation, $\mu > \kappa\,\eta_j$, then the decision on $\hat{A}_\ell$ is *provably identical* to the decision on the full dense tensor A_ℓ — no accuracy loss, only earlier stopping. Communication then scales as

$$(\text{factor size for } L_\ell) \;+\; O\big(k\log(N/k)\big) \quad \text{instead of } O(N),$$

with k the number of sparse blocks actually requested before certification.

Target Theorem 1 (to be proved)**.** *Under the $A = L + S + E$ model with $\|E_\ell\| \le \varepsilon_0$ and a κ-Lipschitz downstream map, the progressive protocol returns a decision identical to dense inference on every input whose margin exceeds $\kappa(\varepsilon_0 +$ residual after k blocks$)$, using expected communication $O(r\cdot$(mode size)$+\bar{k}\log(N/\bar{k}))$, where $\bar{k}$ is the mean certified sparse budget over the input distribution.*

3.5 Communication scaling — the payoff

For dense transmission, cost $= O(N)$. For LoSpar-CI, cost $=$ (factors) $+ O(k\log(N/k))$. When A_ℓ is genuinely low-rank-plus-sparse, r and k are tiny relative to N, so the saving grows with ambient dimension — exactly the ultra-high-dimensional regime where dense transmission hurts most.

4 Application domains (why the $A = L + S + E$ premise holds)

- **Hyperspectral imagery — the strongest case.** Low-rank+sparse (+ Gaussian noise) is canonical here: LRTV [37] *literally* decomposes an HSI into low-rank + sparse noise + Gaussian noise (i.e. $L+S+E$); low-rank recovery [38] and low-rank tensor recovery [39] are standard.

- **Dynamic medical volumes.** Otazo et al. [40] decompose accelerated dynamic MRI into low-rank + sparse; CT sequences likewise [41]. *Note:* these exploit *temporal* correlation — for a single static volume the support is spatial compressibility [42] rather than a named $L + S + E$ result.

- **Whole-slide pathology images.** WSIs carry extreme spatial redundancy and are highly compressible [43]. *Honest caveat:* no paper models a WSI literally as $L + S + E$; the justification is high spatial redundancy (a strong L term), so phrase the premise as "low-rank structure with sparse informative regions," validated empirically per modality before relying on certified early-stop.

5 LoSpar-CI for LLMs: extending the FSU pipelined-inference poster

The framework above was developed for ultra-high-dimensional *inputs*. For LLMs, the right object is the *KV cache*, which grows every token. "KV-cache compression" says *what*; "online/real-time" says *when*. The feasible design uses both offline and online work, but assigns them different jobs: a fixed weight-side projection creates the low-rank bulk, while a small input-dependent residual is extracted block-wise as old cache entries leave an exact hot window.

5.1 Why the obvious target (the inter-stage activation) is the wrong one

During autoregressive *decoding* only one token crosses each stage boundary per step, so the transmitted activation is a single hidden vector of size H ($\approx 6\,\text{KB}$ for a 3B model). A lone vector has no low-rank structure to exploit, and the poster's $\approx 34.5\,\text{ms}$/token communication is dominated by *round-trip latency and framework overhead*, not by transmission volume (6 KB over 1 Gbps is $\sim 0.05\,\text{ms}$). Compressing that activation therefore cannot help decoding — an honest non-starter worth stating plainly. (During *prefill* the activation is a genuine (seq $\times$ H) tensor, so progressive low-rank transmission *does* cut long-prompt prefill communication — a secondary win, not the main one.)

5.2 Two routes, and why the hybrid is the feasible one

The earlier draft implicitly used an **activation-side route**: compute full K, V, then repeatedly decompose the growing cache as $K \approx AB^\top + S_K$. This is data-adaptive but puts decomposition on the decoding path. "Real-time" must not mean an SVD every token; on small head dimensions that overhead can exceed dense attention.

The **weight-side route** factorizes W_K, W_V once, offline, so each token is born in a small latent space. Palu [11], Eigen Attention [12], and MLA/ MHA2MLA [14, 15] establish this route. It is LoRA-like in timing (fixed factors used at runtime), but it compresses inference activations rather than representing a trainable weight update.

Central claim

Revised LLM design: use the weight-side route for the low-rank bulk L and the activation-side route only for a sparse innovation S. Keep a recent exact window. When a full block ages out, compute its residual against the fixed latent reconstruction once, retain only the largest blocks/entries, and discard the dense copy. There is no online SVD and no decomposition of the accumulated cache.

5.3 The structural model and direct execution

For an old block B,

$$K_B \approx \hat{K}_B(z_B^K) + S_B^K, \qquad V_B \approx \hat{V}_B(z_B^V) + S_B^V, \qquad z_t^K = x_t U_K,\ z_t^V = x_t U_V.$$

Attention uses the latent term and sparse correction directly; it must not reconstruct the old dense block. This requires a fused kernel and a measured complexity crossover. Eigen Attention already establishes low-rank-space attention, so direct execution alone is not novel.

5.4 RoPE is a design constraint, not a footnote

LLaMA applies RoPE after the key projection, so a generic low-rank basis cannot always be absorbed into the query. The first prototype should preserve complete RoPE coordinate pairs,

following RAP [16]. A higher-compression variant can decouple a small exact positional key from the semantic latent, following MLA/MHA2MLA [14, 15], using calibration plus parameter-efficient tuning. Any reported speedup must include RoPE transforms; Eigen Attention reports that RoPE overhead can erase its latency benefit on LLaMA models.

5.5 Why it fits the poster's architecture

The poster gives each device whole layers and the complete KV for those layers. Each stage can therefore retain its exact hot window plus latent–sparse old cache locally. Remote per-layer KV sharding is intentionally excluded: it would add a network round trip inside every attention layer. Compression budgets and the layer partition are chosen jointly so the slowest stage, not average work, is minimized.

5.6 The certificate becomes a budget allocator, not a per-token loop

A global logit certificate that recomputes downstream layers after every residual increment is too expensive for the fast path. Instead, calibrate per-layer/head attention-output error $\mathcal{E}_{\ell h}(r, k, b)$ offline and allocate rank, residual blocks, and bit-width under a global quality budget. A strict next-token certificate remains a small diagnostic experiment, not the primary runtime protocol. This trades a stronger theorem for a more credible speedup.

5.7 Which poster problems it helps — and which it does not

Poster problem	Helps?	Mechanism
Memory wall (its core problem)	**yes, directly**	Latent+sparse old KV shrinks the per-device footprint without online SVD.
Context-length limit (its *next* wall)	**yes**	KV grows with conversation; compressing it pushes the wall out ⇒ longer chats on the same 3 devices.
Layer partition / load balancing	**yes**	A smaller KV budget relaxes the 16/6/6 memory constraint ⇒ rebalance (where it meets SparsePipe).
Long-prompt prefill communication	partly	Progressive low-rank transmission of the (seq × H) prefill activation.
Per-token decoding communication (∼34.5 ms)	**no**	That is round-trip latency, not size.
Raw per-token compute latency	regime-dependent	A fused latent kernel can help at long context; block aging and indexing can hurt at short context.

Central claim

For LLMs, LoSpar-CI becomes a *RoPE-safe latent–residual KV method for the poster's distributed pipeline*. Its primary claim is a better memory–quality frontier and longer stationary context. Latency is claimed only beyond a measured crossover where fused latent attention repays aging and indexing overhead.

5.8 Positioning versus existing KV compression

Palu/Eigen Attention/MLA already cover weight-side low rank; RAP handles RoPE-aligned reconstruction-free projection; OjaKV [17] combines exact anchors with online subspace adaptation; STAR-KV [18] adapts rank; LCA [19] condenses context directly inside an MLA latent;

and JoLT [20] preserves a projected residual. The defensible novelty hypothesis is therefore the *combination of a fixed RoPE-safe latent, sparse prompt-specific innovations, and joint heterogeneous stage placement with stationary per-layer KV*. The decisive ablation compares joint placement with the same compressor under the fixed 16/6/6 split.

5.9 Experiment plan on the poster's cluster

- **Pre-study:** calibrate fixed projection factors; measure pre/post-RoPE spectra and sparse residual concentration by layer/head. Stop if 2× compression costs more than one percentage point average task quality.
- **Testbed:** the poster's 3× Jetson Orin Nano cluster [3]; compare the fixed 16/6/6 split with the joint planner under identical precision and power modes.
- **Baselines:** dense CI-Inf; Palu, Eigen Attention, RAP, OjaKV, and STAR-KV; KIVI [23]; and paging/offload where portable. The same compressor under a fixed partition is the decisive ablation.
- **Metrics:** KV bytes per token, admitted context length, TTFT, median and P95 token latency, throughput, energy, copy/migration traffic, and task quality.
- **Performance target:** 1.5× end-to-end throughput or 2–4× longer stationary context at matched quality. Treat short-context batch-1 as a negative control.

6 Scoped first paper

Central claim

Fix *one* modality, *one* decomposition, *one* refinement rule, and a small prototype — establish the structural premise empirically before adding scheduling/RL.

- **Modality:** whole-slide images *or* hyperspectral (hyperspectral gives the cleanest $L+S+E$ justification; WSI gives the most compelling scale story).
- **Decomposition:** Tensor-Train for L [8]; block-support encoding for S.
- **Refinement rule:** margin-based certified early stop (§3.4).
- **Prototype:** a two- or three-device edge cluster (as in the poster [3]).
- **Metrics:** communication bytes vs. dense at *certified* (zero) or bounded accuracy loss; fraction of inputs stopping at the coarse stage; end-to-end latency; per-layer $L+S$ energy (a cheap pre-study establishing where the premise holds).
- **Baselines:** dense transmission; a fixed-rate low-rank-only transmission (no sparse refinement); a token-compression method (SparseVLM [26] / DTC-WSI [25]) adapted to the setting.

Main risk & mitigation

The $A = L + S + E$ premise may hold only weakly for some layers/modalities, weakening certified early-stop. Mitigate by measuring per-layer low-rank+sparse energy *first* (a cheap empirical study) and applying LoSpar-CI only where the spectrum justifies it, with dense fallback elsewhere. Second risk: estimating a tight Lipschitz constant κ — use local/empirical estimates and a conservative margin, accepting occasional extra refinement rather than uncertified errors.

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